

More AI Reviewers Don't Help — Decorrelation Does

A pre-registered, compute-matched comparison of four AI code-review architectures across **2,415 reviews** — including 30 real pull requests from the platform we built for our client.

Companion to: “Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study”

En-Ping Su · Tong Wu · Shiting Huang · Mengshan Li

Khoury College of Computer Sciences, Northeastern University · Vancouver · Advisor: Yvonne Coady · Industry Partner: Indigenomics Institute (Carol Anne Hilton, Founder)

THE CONTRIBUTION

Agreement predicts truth — but **only** when the reviewers are independent.

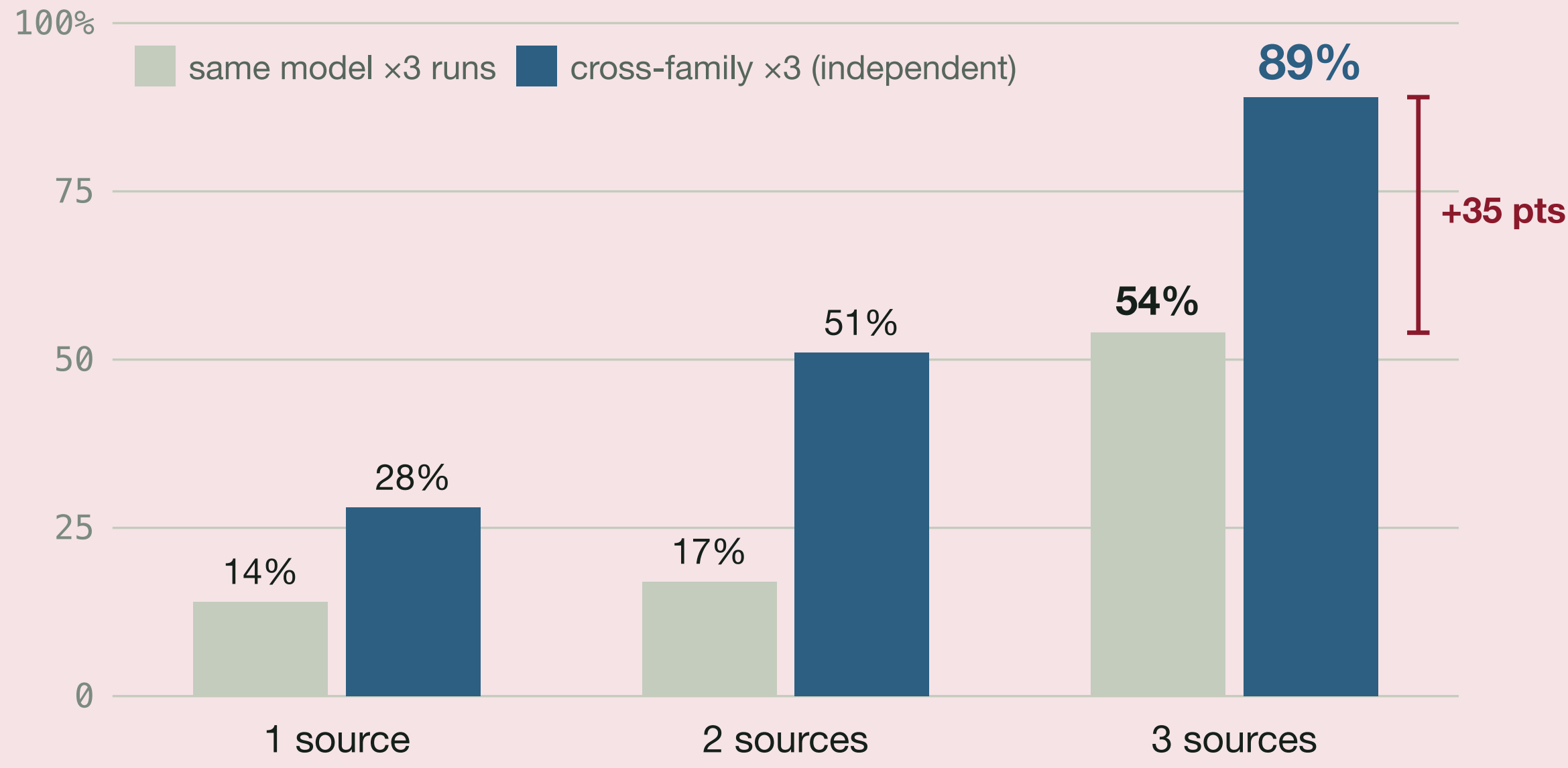
Error decorrelation — not more agents, specialists, or debate — is what turns agreement into a real signal.

89%

three **independent** model families that agree match injected ground truth

54%

one model run three times — agreement is mostly noise



Golden-match rate by number of sources agreeing (E1, injected ground truth).

01 PROBLEM & MOTIVATION

THE CLIENT

The **Indigenomics Institute** needs corporate Reconciliation Action Plan commitments made legible and trackable — and AI now writes much of the software that does it.

THE RESEARCH

Multi-agent AI reviewers are proliferating — managers, specialists, voters — and priced accordingly. Does that **structure** improve review, or just cost more?

GROUNDING IN
Carol Anne Hilton's economic-reconciliation framework — an economic lens, not a case summarizer.

2,415

reviews

179

pull requests

4

architectures

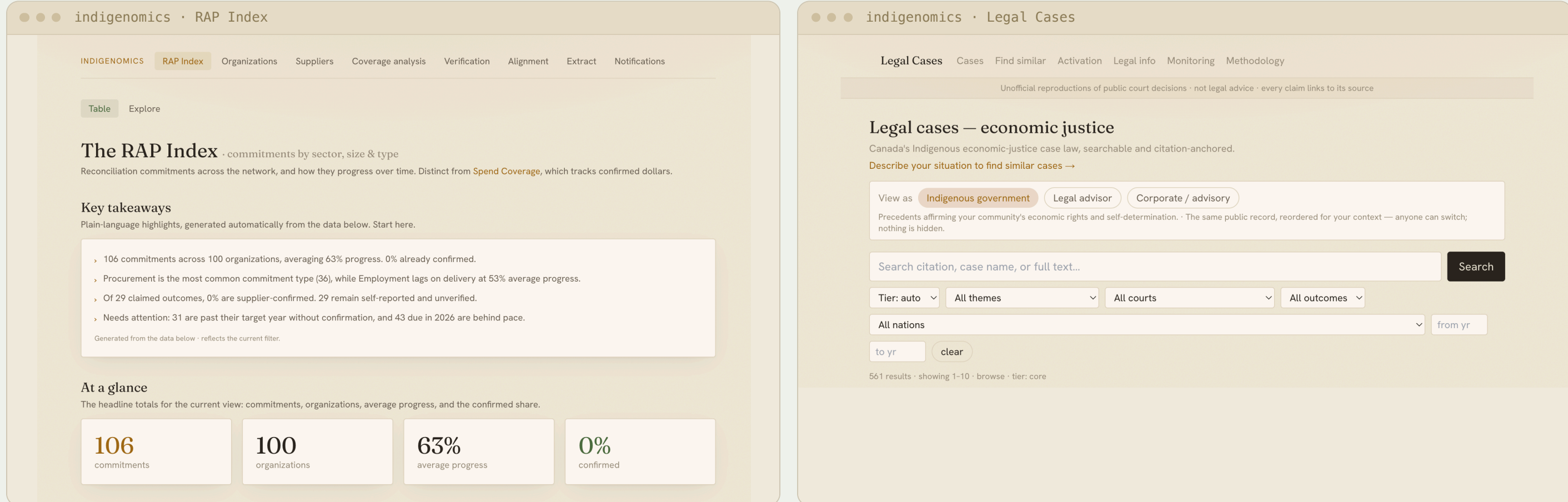
x3

runs each

02 APPROACH

WHAT WE BUILT FOR THE CLIENT

The **RAP Data Portal** — one platform, two lenses: corporate reconciliation commitments and Indigenous legal precedents, as comparable searchable data on AWS.



THE EXPERIMENT — APPARATUS, NOT THE FINDING

Agentless → Generalists-3 → Hierarchical → Consensus

Each rung adds exactly one capability — sampling, then specialization, then communication.

Held constant: same PR snapshot · same model (Claude Haiku 4.5) · same prompts · same evaluation pipeline · three runs per arm · **pre-registered on OSF**

E1 · Qodo

99 PRs · ground truth

E2 · SWE-PRBench

50 PRs · human review

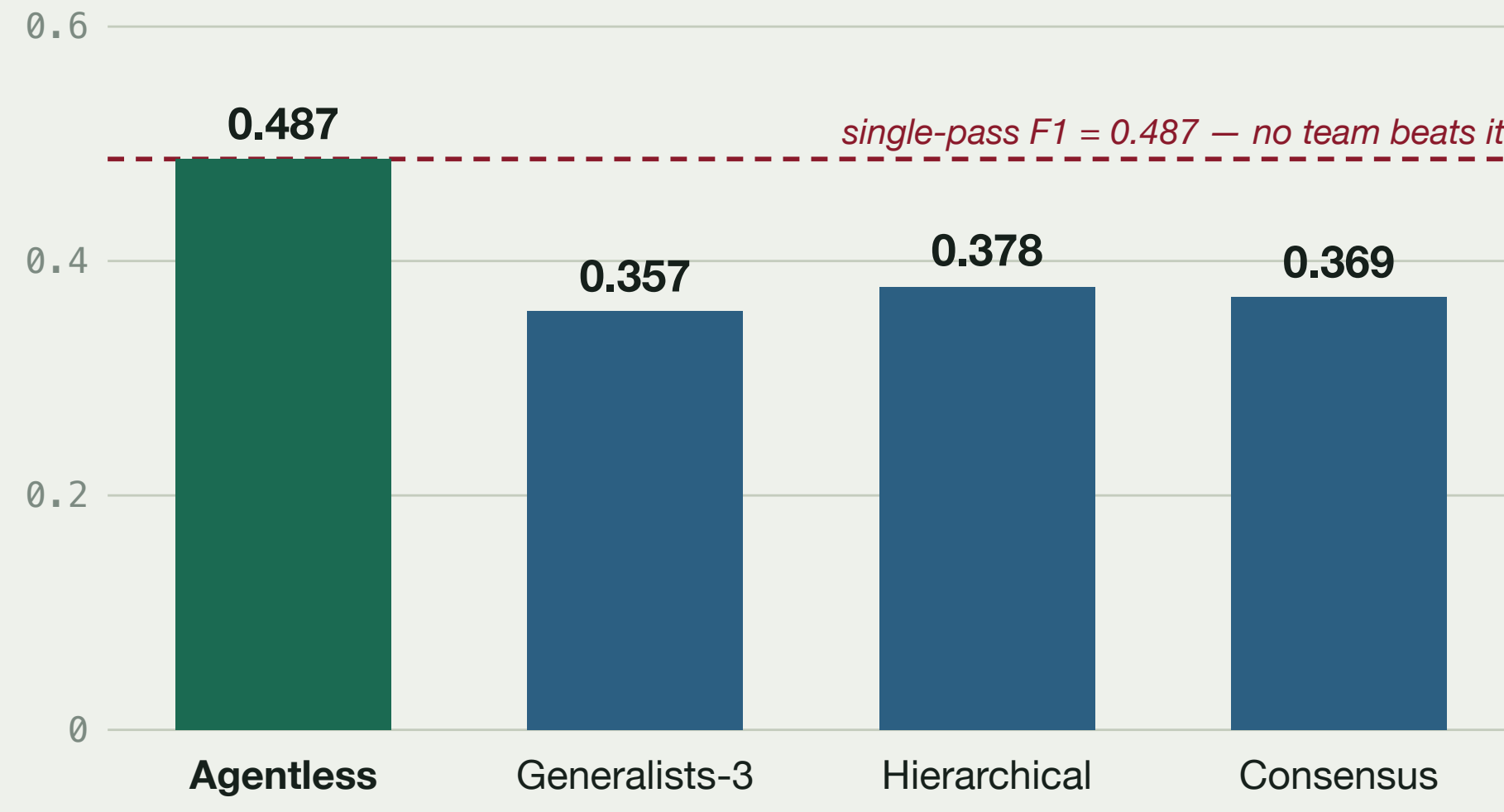
E3 · our portal

30 PRs · no oracle

03 KEY RESULTS

Structure is null. No multi-agent arm beats single-pass F1; specialists tie compute-matched sampling ($p=0.73$); peer debate adds nothing over a manager ($p=0.12$).

Decorrelation pays — on precision, not coverage (89% vs 54%). Bounded: recall plateaus at ≈ 0.83 ; on the portal's own unlabeled code, LLM judges disagree ($\kappa=0.03$).



Semantic F1 by architecture (E1 · Qodo, 99 PRs) — extra agents raise recall but halve precision, so no team beats single-pass.

04 WHAT THE EXTRA AGENTS ACTUALLY BUY · E1 · QODO, 99 PRS



5x more expensive per confirmed finding than single-pass review

Adding agents doesn't buy quality — it buys **recall at a price**. Every confirmed true finding costs **0.34** LLM calls under single-pass review and **1.76** under peer consensus, for a **lower-F1** result.

That is the practical case for decorrelation: **spend the budget on independence, not on more of the same reviewer.**

IMPACT — FOR THE FIELD

Don't deploy multi-agent review expecting a quality lift. **Decorrelate your verifiers** — independent model families — and read their agreement as a precision signal.

IMPACT — FOR THE CLIENT

A verifiable, trackable view of economic reconciliation — **commitments + legal precedents in one platform**, delivered to the Indigenomics Institute.

NEXT STEPS

Tool-grounded verifiers — execution and static analysis as a correctness oracle. **Open science:** all runs, ablations & artifacts released for replay.